TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Total Hardness of Water Using Eriochrome Black T

One reagent that counts calcium and magnesium together — EDTA complexometry at pH 10, a metal-indicator that changes its mind from wine-red to blue, and the $CaCO_3$ convention every hardness figure is reported in.

CHE-1206 · CHEMISTRY-II SESSIONAL · EXPERIMENT 10 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the total hardness of a water sample — the combined Ca^{2+} and Mg^{2+} content — by titration with standard EDTA. EDTA (used as its disodium salt, Na_2H_2Y) is a hexadentate ligand: one EDTA unit wraps a single metal ion in a cage of six donor atoms, whatever the ion's charge:

$$M^{2+} + H_2Y^{2-} \rightarrow MY^{2-} + 2\,H^{+} \quad (M = Ca \text{ or } Mg;\ \text{always } 1 \text{ mol EDTA} : 1 \text{ mol metal ion})$$

That fixed 1 : 1 stoichiometry, with no redox and no precipitate, is what makes the titration clean. Three conditions make it work. **pH 10 buffer** (NH_4Cl/NH_4OH): the reaction releases H^+ and would stall as the solution acidifies, and both the EDTA complexes and the indicator behave correctly only near pH 10. **The indicator, Eriochrome Black T (EBT):** free in solution at pH 10 it is **blue**, but it binds Mg^{2+} (and Ca^{2+}) into a **wine-red** complex — so the sample starts wine-red, and as EDTA strips the metal first from solution and finally from the indicator itself, the last drop frees the dye and the colour snaps to **pure blue**. **Magnesium's quiet role:** the Ca–EBT colour change is weak, so a trace of Mg^{2+} is needed for a sharp end point; most natural waters supply it themselves.

Reporting: hardness is stated as the mass of $CaCO_3$ *equivalent* to the Ca^{2+} + Mg^{2+} present, in mg L^{-1} (ppm) — one common scale for ions of two different masses, since 1 mol of either ion corresponds to 1 mol (100 g) of $CaCO_3$ on the equivalent scale. The scale of the result is read against the usual classification: below 60 mg L^{-1} soft, 60–120 moderately hard, 120–180 hard, above 180 very hard. Hardness is no abstraction in a textile syllabus: it is the Ca^{2+} and Mg^{2+} that waste soap, precipitate dyes and stain fabric in wet processing, and this is the titration the dyehouse laboratory runs on its water supply.

2. Apparatus and solutions

- Standard EDTA solution, 0.0100 M, in the burette.

- Water sample; measuring cylinder or pipette for 50.0 mL.
- pH 10 buffer solution (NH_4Cl/NH_4OH); Eriochrome Black T indicator (solid mixture with NaCl, or solution in ethanol).
- Conical flask, white tile.

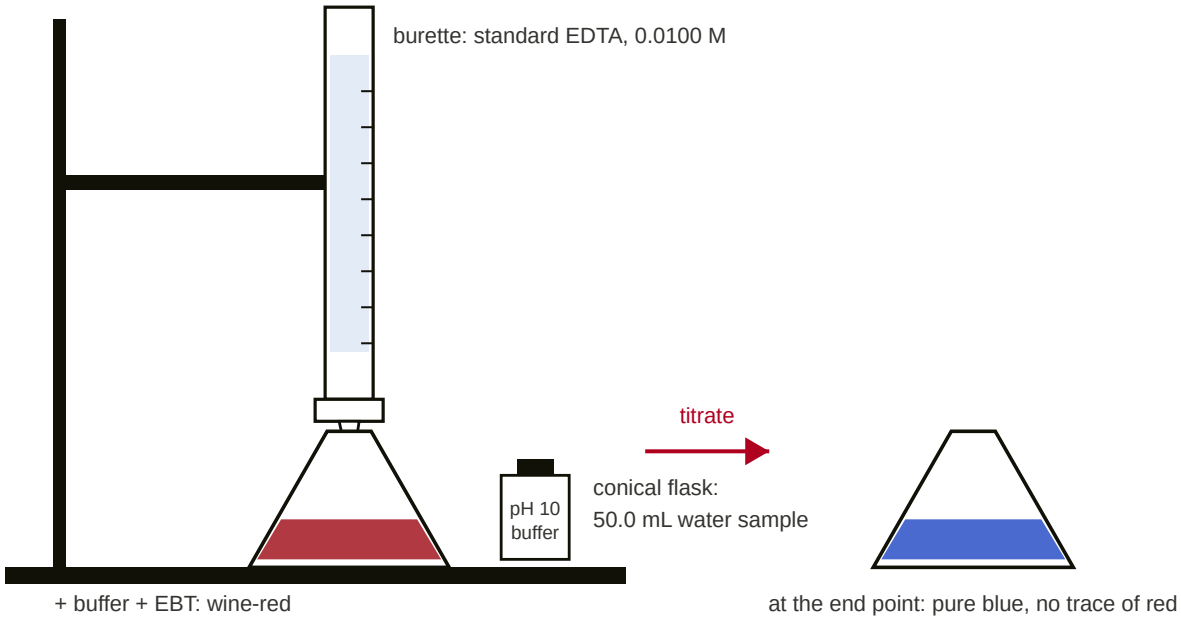


Figure 1 — The hardness titration. EBT holds the sample wine-red while any Ca^{2+}/Mg^{2+} remains for it to bind; the EDTA drop that takes the last metal ion sets the dye free and the flask turns blue.

3. Procedure

1. Rinse and fill the burette with the standard EDTA solution.
2. Measure 50.0 mL of the water sample into a conical flask.
3. Add about 2 mL of the pH 10 buffer, then a pinch (or 2–3 drops) of Eriochrome Black T. The solution turns **wine-red** if hardness ions are present.
4. Titrate with swirling. The colour drifts from wine-red through purple; slow to drops as it does, and stop at the first drop that leaves the solution **pure blue with no reddish tinge**, viewed against the white tile.
5. Repeat until two titres agree within 0.1 mL; average the concordant titres.

4. Worked calculation (model data)

HARDNESS RESULT

Sample volume: 50.0 mL. Standard EDTA: 0.0100 M. Concordant titre: 12.5 mL.

Moles of EDTA = $0.0100 \times 0.0125 = 1.25 \times 10^{-4}$ mol = moles of ($Ca^{2+} + Mg^{2+}$) in the sample (1 :

1).
Equivalent $CaCO_3$ = $1.25 \times 10^{-4} \times 100$ g = 12.5 mg, in 50.0 mL of water.
Total hardness = 12.5 mg × (1000/50.0) = **250 mg L^{-1} as $CaCO_3$** — ‘very hard’ on the usual scale (above 180 mg L^{-1}).
Bench shortcut, worth deriving once: hardness (mg L^{-1}) = titre (mL) × M(EDTA) × 100 000 / sample volume (mL) = 12.5 × 0.0100 × 100 000 / 50.0 = 250.

5. Observation table (as recorded)

TRIAL	INITIAL READING (ML)	FINAL READING (ML)	TITRE (ML)
1 (rough)	0.0	13.0	13.0
2	0.0	12.5	12.5
3	0.0	12.5	12.5

The rough run is doubly useful here: it warns how far the colour has to travel, so the purple in-between stage of the later runs is not mistaken for the end point.

6. Precautions

- Add the buffer before the indicator and titrate without long delay: at pH 10 in open air, $CaCO_3$ can begin to precipitate from a very hard sample and escape the titration.
- Judge the end point against the white tile in good light — the last trace of red in the blue is easy to miss, and one drop too many is a visible error at these titres.
- Use the minimum of indicator: EBT is itself a ligand, and a heavy hand with it consumes EDTA and dulls the colour change.
- Distilled water for rinsing must itself be hardness-free (it is, if truly distilled) — rinsing the flask with tap water would donate the very ions being measured.
- Heavy-metal ions (Cu^{2+}, Fe^{3+}, Al^{3+}) block EBT if present in quantity; in a teaching laboratory the sample is chosen free of them, but a dyehouse effluent would need masking agents first.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is the titration buffered at pH 10? Give both reasons.

2. Describe the colour change of Eriochrome Black T in this titration and explain it in terms of which species holds the dye at the start and at the end.
3. A 50.0 mL sample needs 9.6 mL of 0.0100 M EDTA. Find the total hardness in mg L^{-1} as $CaCO_3$ and classify the water.
4. Why is hardness reported 'as $CaCO_3$' when the ions actually present are Ca^{2+} and Mg^{2+}?
5. Why is the EDTA : metal ratio 1 : 1 for both Ca^{2+} and Mg^{2+} — and why would it still be 1 : 1 for a 3+ ion?
6. A sample containing calcium but essentially no magnesium gives a sluggish, indistinct end point. Why, and what is the standard remedy?

Answer key

1. The complex-forming reaction releases H^+ ($M^{2+} + H_2Y^{2-} \rightarrow MY^{2-} + 2\ H^+$), so without a buffer the pH would fall, the conditional stability of the complexes would collapse, and the titration would stop short. Also, EBT shows its blue free colour and wine-red metal colour in the right forms only near pH 10 — the indicator itself is pH-sensitive.
2. At the start the dye is bound to Mg^{2+}/Ca^{2+} as a wine-red metal–indicator complex. EDTA first complexes the free metal ions, then — being the stronger ligand — strips the metal from the dye at the end point; the liberated dye at pH 10 is blue. Wine-red → blue.
3. Moles EDTA = $0.0100 \times 0.0096 = 9.6 \times 10^{-5}$ = moles of hardness ions; $CaCO_3$ equivalent = $9.6 \times 10^{-5} \times 100$ g = 9.6 mg in 50.0 mL → 192 mg L^{-1} — very hard (above 180).
4. Because the titration counts moles of divalent ions without distinguishing them. Expressing the total as the mass of $CaCO_3$ carrying the same number of equivalents gives one agreed scale on which any mixture of Ca^{2+} and Mg^{2+} can be reported and compared — a convention, not a claim that $CaCO_3$ is in the water.
5. EDTA is hexadentate: a single EDTA unit supplies all six donor atoms around one metal ion, so one EDTA saturates one ion regardless of its charge — 1 : 1 for Ca^{2+}, Mg^{2+}, and equally for Fe^{3+} or Al^{3+}. Charge affects the complex's net charge, not the count.
6. EBT's colour change is sharp with Mg^{2+} but weak with Ca^{2+} alone, so the red-to-blue transition drags. The remedy is to add a little Mg–EDTA complex to the flask: it releases Mg^{2+} to the indicator while consuming no net EDTA, restoring a crisp end point.
